

ANDRY HUANG

Senior Fullstack & AI Engineer (.NET | Next.js | Node.js | RAG / AI Agents)

Surabaya, Indonesia | +62 813-5729-6386 | andrych17@gmail.com
LinkedIn: [linkedin.com/in/andry-huang-ba410a170](https://www.linkedin.com/in/andry-huang-ba410a170) | Website: <https://andryhuang.com>



PROFESSIONAL SUMMARY

Senior Fullstack & AI Engineer with 7+ years of experience architecting and building enterprise SaaS platforms, distributed systems, and production AI tools. Core stack: .NET Core, Next.js, Node.js, NestJS, SQL performance tuning, and LLM RAG pipelines. Proven background delivering secure, high-concurrency architectures with international remote engineering teams.

TECHNICAL SKILLS

- **Backend & Cloud Architecture:** .NET Core, C#, Node.js, NestJS, Laravel, Java Spring Boot, Python (FastAPI), REST & GraphQL APIs, SaaS Architecture
- **AI Engineering & Automation:** RAG Architecture, OpenAI GPT-4o, Anthropic Claude API, Model Context Protocol (MCP), Autonomous AI Agents, pgvector, BullMQ, Prompt Engineering
- **Frontend & Mobile:** Next.js, React, TypeScript, JavaScript, Tailwind CSS, Flutter, Android Native, Livewire, Alpine.js
- **Databases & Storage:** PostgreSQL, SQL Server, MySQL, Oracle DB, Redis, Elasticsearch, AWS S3, Azure Blob, Cloudflare R2
- **DevOps, CI/CD & Security:** Docker, GitHub Actions, Azure DevOps, GitLab CI, Linux Administration, Nginx, Fail2ban, SonarQube, SSH Security
- **Testing & Integrations:** Playwright, Selenium, Midtrans Payment Gateway, RajaOngkir, RFID Hardware, Leaflet GIS

WORK EXPERIENCE

Fullstack Developer | MRI Software (formerly Anacle Systems Ltd) (Singapore (Remote)) | Nov 2022 - Mar 2026

- Scaled enterprise SaaS platforms (Anacle Simplicity & SMRT Tenant Management System), supporting mall tender submissions, space leasing, tenant applications, and day-to-day operations using .NET Core, .NET Framework 4.8, Next.js, and NestJS.
- Deployed production RAG pipelines using OpenAI API and PostgreSQL (pgvector) for semantic document retrieval, search, and automated executive reporting.
- Built a document migration microservice syncing 100,000+ enterprise files to SharePoint Graph API, AWS S3, and Azure Blob Storage with RBAC security and audit logging.
- Refactored legacy .NET stored procedures and wrote automated regression tests using Selenium & Playwright across core SaaS modules.
- Led SonarQube static code analysis across enterprise repositories, fixed security vulnerabilities, configured CI/CD quality gates, and enforced zero-vulnerability deployment standards.

Fullstack Developer | Software House (Surabaya, Indonesia) | Mar 2022 - Nov 2022

- Built high-performance web applications using Node.js and Next.js, optimizing complex SQL queries for high-concurrency transactions.
- Recovered and repaired 10,000+ corrupted database records by identifying and patching edge-case business logic bugs.
- Maintained self-hosted Git servers, configured Elasticsearch clusters for log indexing, and upgraded enterprise Java runtimes.
- Resolved recurring high-priority incidents by diagnosing and fixing core backend bottlenecks.

Full-Stack Developer | PT Pabrik Kertas Tjiwi Kimia Tbk (Surabaya / Mojokerto, Indonesia) | Sep 2019 - Feb 2022

- Developed manufacturing dashboards and reporting systems for pulp & paper production using .NET Core, C#, ASP Classic, and SQL Server.
- Built an automated WhatsApp alert bot using Python, Java, and Selenium, capturing real-time production metrics for plant OEE and HWT targets.
- Designed and deployed a native Android Barcode Scanner application for warehouse inventory tracking.
- Delivered enterprise internal applications including an automated COVID-19 employee health check-in platform and digital corporate raffle draw systems.

FEATURED PROJECTS & INNOVATIONS

- **Qualiv | AI Recruitment SaaS Platform** (Founder & Lead Architect): Built a multi-tenant AI recruitment SaaS featuring automated CV parsing (PDF/DOCX), candidate logic testing, async video interviews, and candidate scoring engine. Tech Stack: Next.js, NestJS, OpenAI GPT-4o, PostgreSQL, Prisma, BullMQ, Redis Queue, Cloudflare R2, Midtrans
- **OpenClaw | AI VPS Security & Ops Agent** (Autonomous AI Agent): Linux VPS monitoring and security agent featuring LLM log anomaly diagnosis, real-time SSH brute-force detection, Fail2ban auto-mitigation, and Telegram alerts. Tech Stack: Node.js, OpenAI API, Telegram Bot API, Fail2ban, Docker, Linux Security
- **PT Nagamasban | AI Sales Assistant & Catalog** (Lead AI Architect): AI recommendation engine and sales training tool for a Surabaya tire distributor. Matches tire specifications with vehicle models and customer budgets using OpenAI GPT-4o embeddings and semantic search. Tech Stack: Next.js, NestJS, OpenAI API, Vector Embeddings, Tailwind CSS
- **AI Portfolio Assistant & Interactive Chatbot** (Creator): RAG-powered portfolio chatbot allowing recruiters and visitors to query professional background, tech stack, and system architecture in real time. Tech Stack: Next.js, OpenAI API, Vector Embeddings, Tailwind CSS
- **Retail & Enterprise Custom ERP / POS Systems** (Fullstack Architect): Delivered custom solutions for retail enterprises: (1) Cahaya Terang Tire ERP (Double-entry A/R & A/P Ledger), (2) Wijaya Mas Jewelry POS (RFID Hardware integration), (3) Knit and Cro E-Commerce (Midtrans & RajaOngkir API), and (4) Tiga Putra Grocery POS (FIFO Expiry tracking). Tech Stack: Laravel, Livewire, MySQL, RFID Integration, Payment Gateway
- **CG Home Sharing & Interactive Quiz Platforms** (Community Initiative Lead): Built community platforms including an Airbnb-style room booking system with Leaflet GIS maps and surveyor inspection workflows, and a real-time WebSocket online quiz and logic testing platform. Tech Stack: Next.js, NestJS, PostgreSQL, Prisma, Leaflet GIS, WebSockets

EDUCATION

Bachelor of Science in Computer Science (S1 Teknik Informatika) | University of Surabaya (UBAYA), Indonesia (2015 - 2019) | GPA: 3.60 / 4.00 (Cum Laude)

CERTIFICATIONS

- **Anthropic AI Certifications Suite (9 Certifications)** (Anthropic, 2026): **Building with Claude API** | **Claude Code in Action** | **Intro to MCP** | **MCP Advanced** | **Agent Skills** | **Subagents** | **AI Capabilities & Limitations** | **Claude Cowork** | **Claude 101**
- **AI Agents with Model Context Protocol: Vanderbilt University** (**Coursera Verification**, 2026). Hands-on implementation of autonomous agent architecture, tool use, and MCP server-client protocols.
- **Software Architecture & Design of Modern Large Scale Systems: Udemy** (**Udemy Verification**, 2026). High-throughput distributed systems, microservices design, caching strategies, and fault-tolerant architecture.

LANGUAGES

- English (Professional Working Proficiency), Indonesian (Native Proficiency)